now. We can't have a commercial deployment of sodium-ion for another 5-10 years. Though we always try to find initial applications for this, thinking about energy storage, mobility, and bigger amounts of deployments, sodium-ion is twenty years away from maturity," says Gupta. "So, the only chemistry, a bit expensive and significantly safer for Indian conditions, is currently the LFP," he adds.

Battery charging

Limiting the use of NMC. Meanwhile, Gupta suggests banning NMC batteries in India. NMC batteries are cheap with high energy. But these are considered primarily responsible for almost all the accidents due to substandard electronic control. "Not because it has bad chemistry but, as a country, we do not know how to use it. So, unless we are able to significantly improve compliance in India, we should limit its use to avoid accidents," says Gupta.

Unfortunately, experts have seen a tendency in opting for cheaper battery solutions rather than safety. "It is difficult to work with such customers or vendors," Gupta says.

Battery engineering. The world has benefitted from the NMCs. But, at the same time, it is a result of a lot of effort in designing the battery. The intelligence is at the NMC integration level, not in the cells. "Any chemistry with appropriate engineering is good," Patwardhan says.

Patwardhan puts emphasis on integration of NMCs as the reason behind that kind of battery pack with energy and cost efficiency. He explains, "The first interaction happens from anywhere in dollars per KW. Ultimately, economics of business is going to drive what you want to put into that particular application. But safety cannot be the least priority."

The qualification of the batteries in Indian standards has to look beyond homologation and mandation as a basic hygiene of the product. Patwardhan says, "Almost all batteries pass, and they don't run hazardous even at the nascent stage of the deployment. So, there is also a need to look at it holistically with regards to homologation and awareness when we put it into the market."

Gupta also highlights that it is not about chemistry, but how well you are able to engineer it. He says, "I am a firm believer in standards, but I am a much firmer believer in the importance of compliance of the standards. So, all the standards may be very good, but compliance is extremely weak. And if you have such weak compliance, the only way out of it is just to ban it."

Is recycling practical?

A hot topic among batteries is recycling. There are companies that work on taking out raw material from lithium-ion cells for repurposing it. There is a huge economics that drives this entire cell recycling scenario, but from a manufacturer's side, it would be called cell management or cell disposal, or something like e-waste management.

Naskar has witnessed many companies struggling to take care of e-waste produced by lithium-ion cells. There will always be rejections from your production and there is usually confusion on where to keep or dump it. He says, "I have seen a huge gap and the vendors have also yet not been able to really deliver on what they wanted to do with those cells lost in our production or testing."

Naskar says, "I think the minimum requirement for a company to take up and make a profit out of it is 500kg or one tonne worth of bad cells, which can be recycled. So, accumulating that amount of waste for management will itself be a problem in most facilities. It's not a very straightforward thing, but there are a lot of stakeholders in the entire process." He adds, "It is a chicken and egg problem, because we don't have that much production capacity in the country. The efficiency of recycling really depends on the process used."

Kumar feels recycling batteries will not help him sell more robots in India, but it may help sell outside India. He says, "Sustainability is something that is given a premium in more advanced markets like European or Japanese. So, it can be feasible for markets abroad."

Charging vs swapping

There are some serious talks on battery charging infrastructure, fast charging capabilities, and even swappable batteries, all of which have a role to play in the large geography of India. Gupta says, "My company specialises in swapping, but we respect charging, home charging, and fast charging, and we believe a combination of the three exists in the market."

With technologies in the nascent stage, the ultimate aim of the country is to have its users trained and free from anxiety. If a certain battery can be charged fast, then yes. If it can be done by swapping, then yes. But both have different pros and cons, and would exclusively co-exist in certain pockets of business segments. Patwardhan says this will evolve with time, but both these should be looked at as energy re-